Sri Chaitanya Educational Institutions | Infinity Learn

Sri Chaitanya Techno School
The right mentor for IIT (JEE), NEET, Olympiad & all Other Competitive exams

41 Years of Excellence 1986-2026

PERFECT SCORE.

Excellence has a permanent address - Sri Chaitanya.

JEE Main 2026 (Session 1)



300/300

PASALA MOHITH

Appl.No: 260310432915

CLASSROOM STUDENT

Congratulations To Parents, Teachers & Staff

#TransformingYourDreamsIntoReality

www.srichaitanya.net

040-66 06 06 06

For More Join: @IITJEE_Advanced

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN-Shift-1** **Date: 31-03-2026**

Time: 09.00Am to 12.00Pm **GTM-34** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

Sri Chaitanya Educational Institutions
Infinity Learn by Sri Chaitanya
41 1986-2026
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 SINGARAJU VENKAT KOUNDINYA
JEE Advanced 2023 RANK 1 VAVILALA CHIDVILAS REDDY
NEET 2023 RANK 1 BORA VARUN CHAKRAVARTHI
For More Join: @IITJEE_Advanced

31-03-2026_ Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main_Shift-1_GTM-34_Test Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Rankers Academy JEE

Sri Chaitanya Educational Institutions

Infinity Learn



For More Join: @IITJEE_Advanced

MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Sri chaitanya prathibha campus whishing you all "HAPPY SANKRANTI" . The number of 4 letter words can be formed by using the letter of the word "HAPPY SANKRANTI" is a 4-digit number $\alpha\beta 90$ and A is a 3×3 matrix such that $|A|=\beta+\alpha-6$, B = adj A then $|A^{-1}.(AB)^T|=$ ____

1) 29 **2)** 18 **3)** 9 **4)** 49

2. $\int\left(\frac{1}{x}+\frac{1}{x^3}\right)\left(\sqrt[23]{3x^{-24}+x^{-26}}\right)dx=\frac{-\alpha}{3(\alpha+1)}\left(3x^{\beta}+x^{\gamma}\right)^{\frac{\alpha+1}{\alpha}}+C,\ x>0,\alpha,\beta,\gamma\in Z$ where C is the constant of integration and $\alpha+\beta+\gamma=\lambda$ then the number of natural numbers less than λ and co-prime to it _____

1) 9 **2)** 19 **3)** 20 **4)** 18

3. Let z_1 and z_2 be two complex number such that $|z_1|=1$ and $|z_2|=10$.If $\theta=\arg\left(\frac{z_1-z_2}{z_2}\right)$, and maximum value of $\tan^2\theta$ is $\frac{1}{\lambda}$ then sum of digit of the λ

1) 99 **2)** 20 **3)** 18 **4)** 8

4. An ellipse $E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,(a<b)$ passes through the vertices of the hyperbola $H:\frac{x^2}{49}-\frac{y^2}{64}=-1$.Let the major and minor axes of the ellipse E coincide with the transverse and conjugate axes of the hyperbola H. Let the product of the eccentricities of E and H be $\frac{1}{2}$. If l is the latus rectum of the ellipse E, then the value of $113l$ is equal to

1)1556 **2)**1552 **3)**1662 **4)**776

Sri Chaitanya Educational Institutions Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 JEE Advanced 2023 RANK 1 NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

5. Let $\bar{\alpha}=4\bar{i}+3\bar{j}+5\bar{k}$ and $\bar{\beta}=\bar{i}+2\bar{j}-4\bar{k}$ .let $\bar{\beta_1}$ be parallel to $\bar{\alpha}$ and $\bar{\beta_2}$ be perpendicular to $\bar{\alpha}$ .If $\bar{\beta}=\bar{\beta_1}+\bar{\beta_2}$ then $5\left(\bar{\beta_2}-\bar{\beta_1}\right)=$_____

1) $13\bar{i}+16\bar{j}-10\bar{k}$ **2)** $13\bar{i}-16\bar{j}+10\bar{k}$

3) $-13\bar{i}+16\bar{j}+10\bar{k}$ **4)** $13\bar{i}-10\bar{j}+16\bar{k}$

6. Let f be a differentiable function defined in [0,1] such that $f(f(x))=x$ and $f(0)=1$.if the value $\int_0^1 (x-f(x))^{2026}dx=\frac{p}{q}$ where p and q are relatively prime then the value of $(p+q)$, is_____

1)2023 **2)**2024 **3)**2028 **4)**2022

7. Statement I: If $a_1x^2+b_1x+c_1=0$ and $a_2x^2+b_2x+c_2=0(a_1\neq 0,a_2\neq 0)$ have a common root. Then $(c_1a_2-c_2a_1)^2=(b_1c_2-b_2c_1)(a_1b_2-a_2b_1)$

Statement II: The quadratic equations $x^2-6x+a=0$ and $x^2-cx+6=0$ have one root in common. The other roots of the first and second equations are integers in the ratio 4:3, If the common root is λ then λ^3 is 8

1) Both Statement-I and Statement-II are correct
2) Only Statement-I is correct
3) Both Statement-I and Statement-II are incorrect
4) Only Statement-II is correct

8. The 4th term of GP is 500 and its common ratio is $\frac{1}{m}, m\in N$ .Let S_n denote the sum of the first n terms of this GP. If $S_6>S_5+1$ and $S_7<S_6+\frac{1}{2}$, and the number of possible values of m is λ. Then the variance of observations $\lambda-11,\lambda-10,\lambda-9,....\lambda-1=$

1)12 **2)**14 **3)**10 **4)**16

Sri Chaitanya Educational Institutions Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 JEE Advanced 2023 RANK 1 NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

9. Four tickets are marked 00,01,10 and 11 respectively, are placed in a bag. A ticket is drawn at random five times, being replaced each time. The probability that the sum of the numbers on the five tickets drawn is 24, is $\frac{2^{\lambda}+9}{512}$. One of the solution of the equation $\tan^2 x = \alpha in\left[0,\frac{\pi}{2}\right)$ where α is the unit digit of $\left(1!+2!+3!+....+\lambda!\right)^{2026!}$

1) $\frac{\pi}{3}$ **2)** $\frac{\pi}{4}$ **3)** $\frac{\pi}{10}$ **4)** $\frac{\pi}{12}$

10. If α^{th},β^{th} terms are independent of x in the expansion of $\left(3x-\frac{1}{x}\right)^{20}$ and $\left(x+\frac{\sqrt[9]{3^{10}}}{x}\right)^{18}$ respectively then number of prime factors of $(\alpha+\beta)^{2026} = ___$

1)1 **2)** 2 **3)** 3 **4)** 4

11. Let $x=2t, y=\frac{t^2}{3}$ be a conic. Let S be the focus and B be the point on the axis of the conic such that $SA \perp BA,$ where A is any point on the conic. If k is the ordinate of the centroid of $\Delta SAB,$ and $\lim_{t\to 1} k$ is equal to λ then $\left[\frac{\lambda}{1}\right]+\left[\frac{\lambda}{2}\right]+....\left[\frac{\lambda}{2026}\right] = ___$ (where [.] GIF)

1) 0 **2)** 1 **3)** 2026 **4)** 2024

12. Let $f(x)$ be a continuously differentiable function on the interval $(0,\infty)$ such that $f(1)=2$ and $\lim_{t\to x}\frac{t^{10}f(x)-x^{10}f(t)}{t^9-x^9}=1$, for all x>0, $f(x)$ is equal to

1) $\frac{31}{11x}-\frac{9}{11}x^{10}$ **2)** $\frac{9}{11x}+\frac{13}{11}x^{10}$ **3)** $\frac{-9}{11x}+\frac{31}{11}x^{10}$ **4)** $\frac{13}{11x}+\frac{9}{11}x^{10}$

13. Let the values of λ for which the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-\lambda}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ is $\frac{1}{\sqrt{6}}$ be λ_1 and λ_2 .then the radius of the circle passing through points $(0,0),(\lambda_1,\lambda_2)$ and (λ_2,λ_1) is

1) $\frac{5\sqrt{2}}{3}$ **2)**4 **3)** $\frac{\sqrt{2}}{3}$ **4)**3

Sri Chaitanya Educational Institutions Infinity Learn


For More Join: @IITJEE_Advanced

14. A circle C_1 passes through the origin O and has diameter 4 on the positive x-axis. The line $y=2x$ gives a chord OA of a circle C_1.Let C_2 be the circle with OA as a diameter. If the tangent to C_2 at the point A meets the x-axis at P and y-axis at Q. Then QA: AP is equal to:

1) 1:4 **2)** 1:5 **3)** 2:5 **4)** 1:3

15. Let f and g be functions satisfying $f(x+y)=f(x)f(y), f(1)=7$ and $g(x+y)=g(xy), g(1)=1$, for all $x,y\in N$. If $\sum_{x=1}^{n}\left(\frac{f(x)}{g(x)}\right)=19607$, then n is equal to

1) 7 **2)** 6 **3)** 4 **4)** 5

16. The interval in which the function $f(x)=x^x\,(x>0)$ is strictly increasing is____

1) $\left(0,\frac{1}{e}\right]$ **2)** $(0,\infty)$ **3)** $\left(\frac{1}{e},\infty\right)$ **4)** $\left(\frac{1}{e^2},1\right)$

17. Let $\alpha,\beta,(\alpha<\beta)$, be the solutions of the equation

$$\sum_{n=1}^{6}\frac{1}{\sin\left(\theta+(n-1)\frac{\pi}{4}\right)\sin\left(\theta+\frac{n\pi}{4}\right)}=4\sqrt{2} \text{ for } 0<\theta<\frac{\pi}{2} \text{ then } |\alpha-\beta|=____$$

1) 0 **2)** $\frac{3\pi}{2}$ **3)** $\frac{\pi}{3}$ **4)** $\frac{\pi}{6}$

18. The set of all x satisfying the equation $x^{\log_3 x^2+(\log_3 x)^2-10}=1/x^2$ is $(x>1)$

1) $\{9\}$ **2)** $\{9,1/81\}$ **3)** $\{4,1/81\}$ **4)** $\{9,1/8\}$

19. The median of a set of 9 distinct observations is 20.5.If each of the largest 4 observations of the set is increased by 2 then the median of the new set

1) Is increased by 2 **2)** Is decreased by 2

3) Is two times the original median **4)** Remains the same as that of original set

20. Bag A contains 3 white,7 red balls and Bag B contains 3 white,2 red balls. One bag is selected at random and a ball is drawn from it. The probability of drawing the ball from the bag A, if the ball drawn is white, is

1) 1/4 **2)** 1/9 **3)** 1/3 **4)** 2/10

Sri Chaitanya Educational Institutions Infinity Learn 41



For More Join: @IITJEE_Advanced

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. A, B, C are the vertices of triangle with right angled at A and $P(-4,0); Q(0,6)$ are two given points. If the ratio of the distances from each vertex of triangle to P, to that of Q is 2:3 ,the centroid of triangle ABC lies on a circle with radius 'r' then $\left[\frac{r}{\sqrt{13}}\right]$ is equal to _____ ([.] represent GIF)

22. A line 'L' passing through point P(1,4,3) is perpendicular to both lines $\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4}$ and $\frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$. If position vector of point 'Q' on line 'L' is (a,b,c) such that $(PQ)^2=357$, then minimum possible values of a+b+c (which is a multiple of least odd prime)

23. The minimum number of elements that must be added to the relation $R=\{(a,b)(b,c)(b,d)\}$ on the set $\{a,b,c,d\}$ so that it is an equivalence relation is_____

24. The value of $\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$ is $\frac{a}{b}$ then the remainder when $(a+b)^{2026!}$ is divided by 3 =_____(a,b are co-primes)

25. Let the points of intersections of the lines $x-y+1=0, x-2y+3=0$ and $2x-5y+11=0$ are the midpoints of the sides of a triangle ABC. The area of the triangle ABC is 4K and $\int_0^k\left[x^2\right]dx=a-b\sqrt{2}$ then $(a-b)^{2026}=$___ (where [.] mean GIF)

Sri Chaitanya Educational Institutions
Infinity Learn
41


For More Join: @IITJEE_Advanced

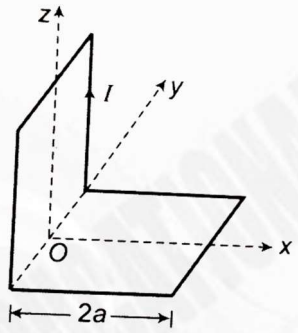
PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

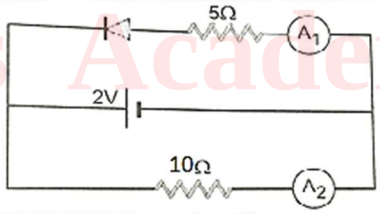
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. A non-planer loop of conducting Wire carrying a current I is placed as shown in the figure. Each of the straight section of the loop is of length 2a. The magnetic field due to this loop at the point $P(a, 0, a)$ points in the direction of ……..



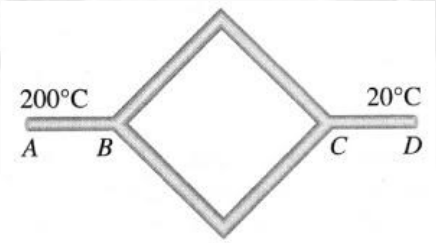
1) $\frac{1}{\sqrt{2}}\left(-\hat{j}+\hat{k}\right)$ **2)** $\frac{1}{\sqrt{3}}\left(-\hat{j}+\hat{k}+\hat{i}\right)$ **3)** $\frac{1}{\sqrt{3}}\left(\hat{i}+\hat{j}+\hat{k}\right)$ **4)** $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{k}\right)$

27. In the following circuit readings in ammeters A_1 and A_2 will be_____



1) 0.2 A, zero **2)** zero, 0.2 A **3)** 0.2 A, 0.2 A **4)** 0.2A, 0.4A

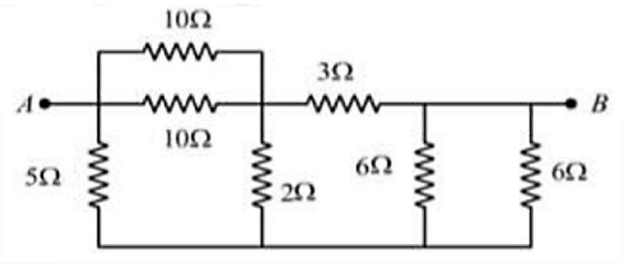
28. Six identical conducting rods are joined as shown in figure. Points *A* and *D* are maintained at temperature 200^0C and 20^0C, respectively. The temperature of junction *B* will be



1) 120^0C **2)** 100^0C **3)** 140^0C **4)** 80^0C

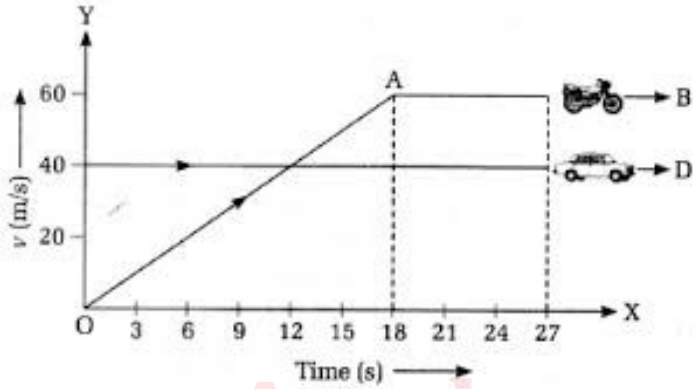
Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 JEE Advanced 2023 RANK 1 NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

29. Find the equivalent resistance between points A and B $(\text{in } \Omega)$



1) 4 **2)** 6 **3)** 11.2 **4)** 10.4

30. At the instant a motor bike start from rest in a given direction, a car overtakes the motor bike, both moving in the same direction. The speed time graphs for motor bike and car are represented by OAB and CD respectively. Then at t = 18s find the distance between the motor bike and car (in m).



1) 60 **2)** 540 **3)** 720 **4)** 180

31. In the relation $P=\frac{\alpha}{\beta}e^{\frac{-\alpha z}{k\theta}}$ P is pressure, Z is distance, k is Boltzmann constant and θ is the temperature. The dimensions of β will be :

1) $M^0L^2T^0$ **2)** ML^2T **3)** ML^0T^{-1} **4)** $M^0L^2T^{-1}$

32. Statement -1 : Newton's First Law is merely a special case $(a=0)$ of the Second Law.

Statement -2 : Newtons's First Law defines the frame from where Newton's Second Law; $\vec{F}=m\vec{a}$, $\vec{F}$ representing the net real force acting on a body, is applicable.

1) If both statements are true and statement 2 is the correct explanation of statement 1.

2) If both statement are true but statement 2 is not the correct explanation of statement 1.

3) If statement 1 is true and statement 2 is false

4) If statement 1 is false but statement 2 is true.

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

33. Assertion : A heavy weight is suspended from the spring . A person raises the weight slowly till the spring become slack. The work done by person is W. The energy stored on the stretched spring was E. The gain in gravitational potential energy is $(W+E)$.

Reason : Work done by the spring force is always negative.

1) Assertion and reason are correct and reason is correct explanation of assertion

2) Assertion and reason are correct and reason is not correct explanation of assertion

3) Assertion is false reason is true

4) Assertion is true reason is false

34. Statement 1 : In a hydrogen atom energy of emitted photon corresponding to transition from $n=2$ to $n=1$ is much greater as compared to transition from $n\to\infty$ to $n=2$

Statement-2 : Wavelength of photon is directly proportional to the energy of emitted photon

1) If both statements are true and statement 2 is the correct explanation of statement 1.

2) If both statement are true but statement 2 is not the correct explanation of statement 1.

3) If statement 1 is true and statement 2 is false

4) If statement 1 and statement 2 are false.

35. Assertion : A proton, a deuteron and an $\alpha-$ particle are accelerated by the same potential difference. Their velocities will be in the ratio of $1:1:\sqrt{2}$

Reason : Kinetic energy, $E = qV = \frac{1}{2}m\upsilon^2$

1) Assertion and reason are correct and reason is correct explanation of assertion

2) Assertion and reason are correct and reason is not correct explanation of assertion

3) Assertion is false reason is true

4) Assertion is true reason is false

36. Two coils of self inductance L_1 and L_2 are connected in series combination having mutual inductance of the coils as M. The equivalent self inductance of the combination will be :



1) $\frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{M}$ **2)** L_1+L_2+M **3)** L_1+L_2+2M **4)** L_1+L_2-2M

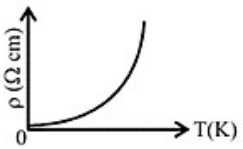
Sri Chaitanya Educational Institutions Infinity Learn

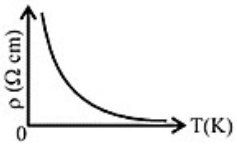


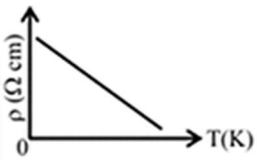
For More Join: @IITJEE_Advanced

37. Which of the following resistivity (ρ) v/s temperature (T) curves is most suitable to be used in wire bound standard resistors?

1) 

2) 

3) 

4) 

38. Match List –I with List –II

List-I	List-II
A .Moment of inertia of solid sphere of radius R about any tangent	I. $\frac{5}{3}MR^2$
B. Moment of inertia of hollow sphere of radius (R) about tangent	II. $\frac{7}{5}MR^2$
C. Moment of inertia of circular ring of radius (R) about its diameter.	III. $\frac{1}{4}MR^2$
D. Moment of inertia of circular disc of radius (R) about any diameter	IV. $\frac{1}{2}MR^2$

Choosen the correct answer from the options given below

1) A-II, B-I, C-IV, D-III **2)** A-I, B-II, C-IV, D-III

3) A-II, B-I,C-III, D-IV **4)** A-I, B-II, C-III, D-IV

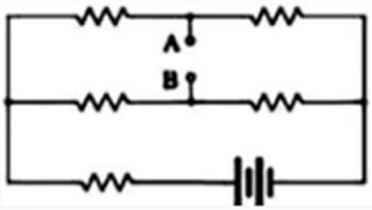
39. An elelctromagnetic wave is represented by the electric field $\vec{E} = E_0\hat{n}\sin\left[\omega t + (6y - 8z)\right]$. Taking unit vectors in x, y and z directions to be $\hat{i}$, $\hat{j}$,$\hat{k}$, the direction of propagation $\hat{s}$, is

1) $\hat{s} = \left(\frac{-3\hat{j} + 4\hat{k}}{5}\right)$ **2)** $\hat{s} = \left(\frac{-3\hat{i} - 4\hat{j}}{5}\right)$ **3)** $\hat{s} = \left(\frac{-4\hat{k} + 3\hat{j}}{5}\right)$ **4)** $\hat{s} = \left(\frac{4\hat{j} - 3\hat{k}}{5}\right)$

40. A network shown in the figure consists of a battery and five unknown resistors. When an ideal ammeter is connected between the terminals A and B, its reading is 4 A and when a resistance of 3Ω is connected in series with the ammeter its reading becomes 2 A. Now the ammeter and the resistance are disconnected and an ideal voltmeter is connected between the terminals A and B. What would the voltmeter read?

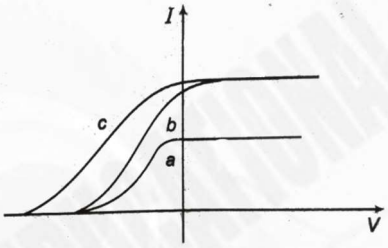
Sri Chaitanya Educational Institutions
Infinity Learn


For More Join: @IITJEE_Advanced



1) 6V **2)** 10V **3)** 12V **4)** 18 V

41. The figure shows the variation of photocurrent (l) with anode potential (V_a) of a photosensitive surface for three different radiations. Let $I_a.I_b$ and I_c be the intensities and $f_a.f_b$ and f_c the frequencies for the waves a, b and c respectively



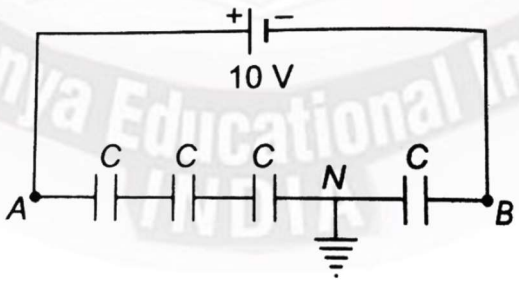
1) $f_a = f_b$ and $I_a \neq I_b$ **2)** $f_a = f_c$ and $I_a = I_c$

3) $f_a = f_b$ and $I_a = I_c$ **4)** $f_b = f_c$ and $I_b = I_c$

42. A diverging lens of focal length 20 cm is placed coaxially 5 cm toward left of a converging mirror of focal length 10 cm. Where would an object be placed toward left of the lens so that a real image is formed on object itself.

1) 30 cm **2)** 40 cm **3)** 60 cm **4)** 80 cm

43. Four identical capacitors are connected in series with a battery of emf 10 V. The point X is earthed. Than the potential of point A is



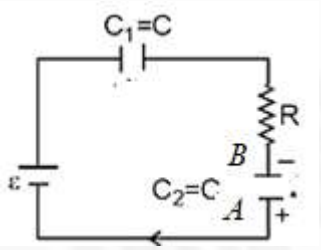
1) 10 V **2)** 7.5 V **3)** -7.5 V **4)** 0 V

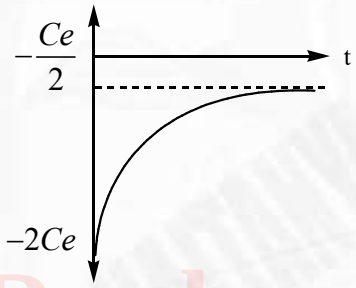
Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1 JEE Advanced 2023 RANK 1 NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

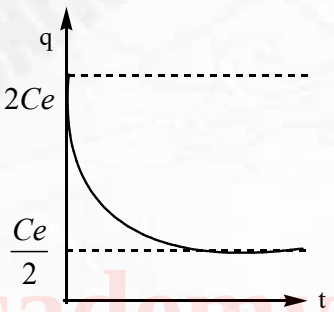
44. A point P lies on the axis of a fixed ring of mass M and radius R, at a distance 2R from its centre O. A small particle starts from P and reaches O under gravitational attraction only. Its speed at O will be

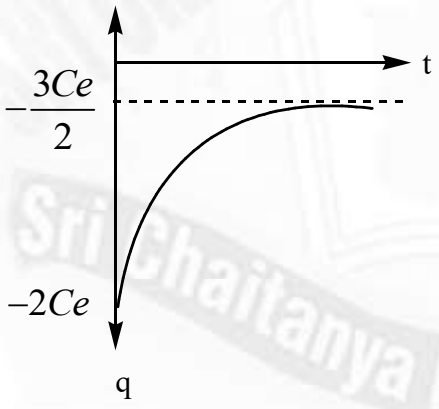
1) zero **2)** $\sqrt{\frac{2GM}{R}}$ **3)** $\sqrt{\frac{2GM}{R}\left(\sqrt{5}-1\right)}$ **4)** $\sqrt{\frac{2GM}{R}\left(1-\frac{1}{\sqrt{5}}\right)}$

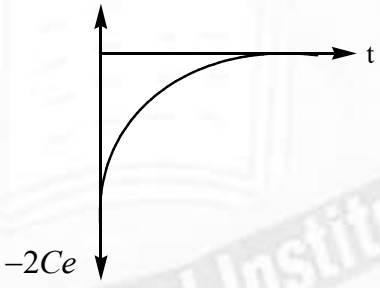
45. The capacitance of C_1 and C_2 shown in diagram has capacitance C each. C_1 is initially uncharged and C_2 is given a charge 2C as shown. Which of the following graph represents charge on plate B of capacitor C_2 as function of time?



1)


2)


3)


4)


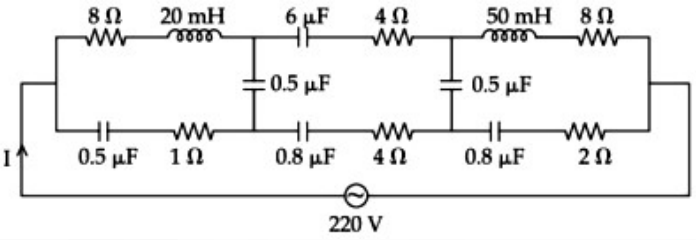
SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

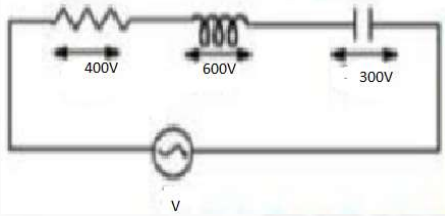
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

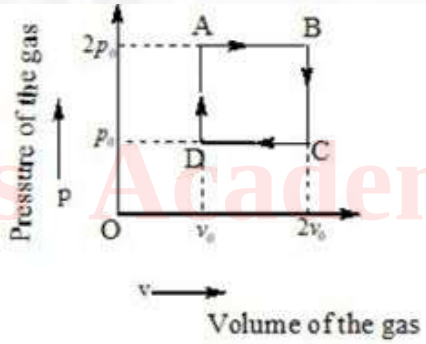
46. The effective current I in the given circuit at very high frequencies will be ____ A.



47. In a series LCR circuit as shown in the diagram. The supply voltage is ____Volt



48. The p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monoatomic gas. The amount of heat extracted from the source in a single cycle is $\left(\frac{x}{2}\right)p_0v_0$. Find the value of x



49. In a Young's double slit experiment, two slits are located 1.5 mm apart. The distance of screen from slits is 2 m and the wavelength of the source is 400 nm, If the 20 maxima of the double slit pattern are contained within the central maximum of the single slit diffraction pattern, then the width of each slit is $x\times10^{-3}$ cm, where x-value is_____

50. A particle performs simple harmonic motion with amplitude A. Its speed is increased to three times at an instant when its displacement is $\frac{2A}{3}$. The new amplitude of motion is $\frac{nA}{3}$. The value of n is_____

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

CHEMISTRY

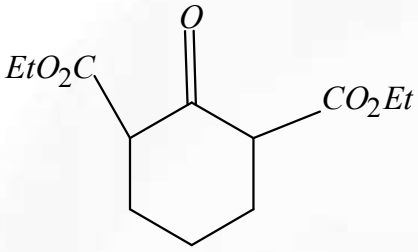
Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

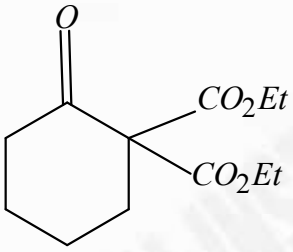
This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

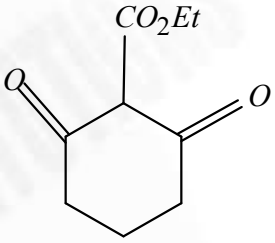
51. Rank in the order of increasing acidic strength



A)



B)



C)

1) $A < B < C$ **2)** $A < C < B$ **3)** $B < A < C$ **4)** $B < C < A$

52. The decomposition of PH_3 on tungsten surface follows the rate equation $\frac{dp}{dt} = \frac{-k^1\ k\ p}{1+kp}$,

(k & k^1 are constants) which of the following statement about the order of the reaction is true

1) First order at all pressures

2) Second order at low pressure and first order at high pressure

3) Zero order at low pressure and first order at high pressure

4) First order at low pressure and zero order at high pressure

53. The pair in which both species have same magnetic moment [spin only] is

1) $\left[Cr(H_2O)_6\right]^{2+}, [CoCl_4]^{2-}$ **2)** $\left[Cr(H_2O)_6\right]^{2+}, \left[Fe(H_2O)_6\right]^{2+}$

3) $\left[Mn(H_2O)_6\right]^{2+}, \left[Cr(H_2O)_6\right]^{2+}$ **4)** $[CoCl_4]^{2-}, \left[Fe(H_2O)_6\right]^{2+}$

54. Choose the correct statement regarding the following complexes

$x = \left[Cr(NH_3)_6\right]^{3+}; y = \left[Ni(CN)_4\right]^{2-}$

1) x is diamagnetic and y is paramagnetic

2) x undergoes d^2sp^3 hybridization and y follows dsp^2 hybridization

3) x is paramagnetic and exhibit sp^3d^2 hybridization

4) y is paramagnetic and exhibits sp^3 hybridization

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

55. Which one of the following is an amphoteric oxide?

1) MnO_2 **2)** ZnO **3)** CaO **4)** CO_2

56. In which of the following pairs, both the species have the same hybridisation?

I) SF_4, XeF_4 II) I_3^-, XeF_2 III) $ICl_4^+, SiCl_4$ IV) ClO_3^-, PO_4^{3-}

1) I, II **2)** II, III **3)** II, IV **4)** I, II, III

57. The reason for the stability of Gd^{3+} ion is

1) $4f$ subshell — half filled

2) $4f$ subshell — completely filled

3) Possesses the general electronic configuration of noble gases

4) $4f$ — subshell empty

58. The increasing order of electron affinity for the following electronic configurations of elements is:

I) $1s^2 2s^2 2p^6 3s^2 3p^5$ II) $1s^2 2s^2 2p^3$ III) $1s^2 2s^2 2p^5$ IV) $1s^2 2s^2 2p^6 3s^1$

1) $II < IV < III < I$ **2)** $I < II < III < IV$ **3)** $I < III < II < IV$ **4)** $IV < III < II < I$

59. 1 litre of 0.1M aq $CuSO_4$ is electrolysed using Pt-electrodes assuming volume of solution to remain constant. Find P^H and volume of gases evolved at STP respectively if 0.4F of electricity is supplied

1) 0.7 & 4.48 lit **2)** 1 & 5 lit **3)** 2 & 3 lit **4)** 3 & 6 lit

60. For the process liquid $\rightarrow$ vapour; liquid at its boiling point and at atmospheric pressure, the correct choice

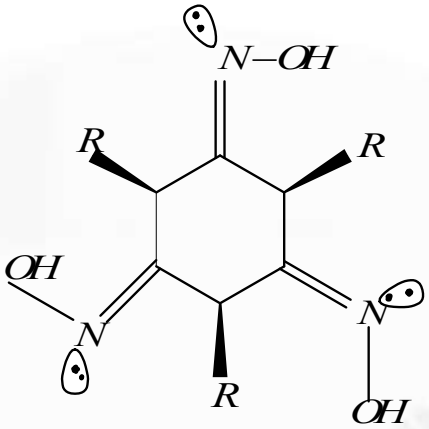
1) $\Delta S_{system} > 0$ *and* $\Delta S_{surroundings} > 0$ **2)** $\Delta S_{system} > 0$ *and* $\Delta S_{surroundings} < 0$

3) $\Delta S_{system} < 0$ *and* $\Delta S_{surroundings} > 0$ **4)** $\Delta S_{system} < 0$ *and* $\Delta S_{surroundings} < 0$

Sri Chaitanya Educational Institutions
Infinity Learn
41


For More Join: @IITJEE_Advanced

61. The given compound has



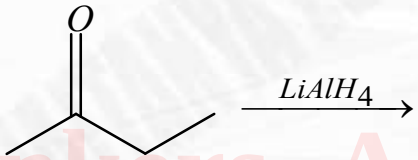
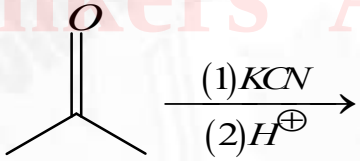
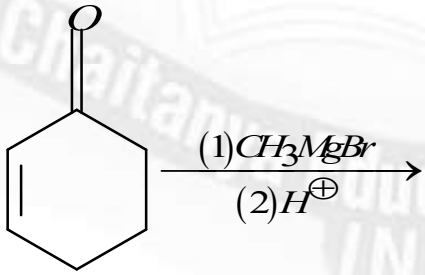
1) Chiral

2) C_3 axis of symmetry

3) Optically active

4) All of these

62. Match the column:

	Column-I		Column-II
a)		p)	Racemic mixture
b)		q)	diastereomers
c)	$Ph-CH_2-Cl \xrightarrow{KCN}$	r)	Nu-addition reaction
d)		s)	Nu-Substitutions reaction

1) $a-r; b-p; c-s; d-s,p$

2) $a-p,r; b-r; c-s; d-p,r$

3) $a-p; b-s; c-r; d-s,p$

4) $a-s; b-p; c-r; d-p,r$

Sri Chaitanya Educational Institutions
Infinity Learn
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

63. The photo-electric emission requires a threshold frequency v_0. For a certain metal $\lambda_1 = 2200\ \overset{0}{A}$ and $\lambda_2 = 1900\ \overset{0}{A}$ produce electrons with maximum kinetic energy KE_1 and KE_2. If $KE_2 = 2KE_1$, v_0 and corresponding λ_0 respectively are?

1) $1.1483\times10^{15}\ \sec^{-1}\ \&\ 2.6126\times10^{-7}\ m$ **2)** $2.6126\times10^{-7}\ \sec^{-1}\ \&\ 1.1483\times10^{15}\ m$

3) $3\times10^{-7}\ \sec^{-1}\ \&\ 2\times10^{-15}\ m$ **4)** $2\times10^{-7}\ \sec^{-1}\ \&\ 3\times10^{-15}\ m$

64. What is the minimum P^H required for the precipitation of Zns in a solution from 0.1 M $ZnCl_2$ saturated with 0.10 M H_2S ?(Given $K_{sp}(Zns) = 10^{-21}, K_{a_1}\times K_{a_2} = 10^{-20}$ for $H_2S, \log 0.316 = -0.5$)

1)1 **2)** 0.5 **3)**2 **4)**2.5

65. Each question contains STATEMENT-I and STATEMENT-II. Each question has 4 choices **(1),(2),(3)** and **(4)** out of which only one is correct marks your answer as

Statement- I: maltose has two β-D-glucose units linked at C_1 and C_4 and is a reducing sugar

Statement- II: maltose has two mono saccharides $\alpha - D$ glucose and $\beta - D$-glucose linked at C_1 *and* C_6 and it is a non-reducing sugar

1) If both statement-I and statement-II are correct

2) If both statement-I and statement-II are incorrect

3) If statement-I is correct but statement-II is incorrect

4) If statement-I is incorrect but statement-II is correct

66. Statement- I: The molecular mass of acetic acid determined by depression in freezing point method in benzene and water was found to be different

Statement- II: Water is polar solvent and benzene is non-polar solvent

1) If both statement-I and statement-II are correct

2) If both statement-I and statement-II are incorrect

3) If statement-I is correct but statement-II is incorrect

4) If statement-I is incorrect but statement-II is correct

Sri Chaitanya Educational Institutions Infinity Learn 41



For More Join: @IITJEE_Advanced

67. Statement- I: The limiting molar conductivity of strong electrolyte ($NaCl$) is higher compared to that of weak electrolyte (CH_3COOH)

Statement- II: molar conductivity decreases with decrease in concentration of electrolyte

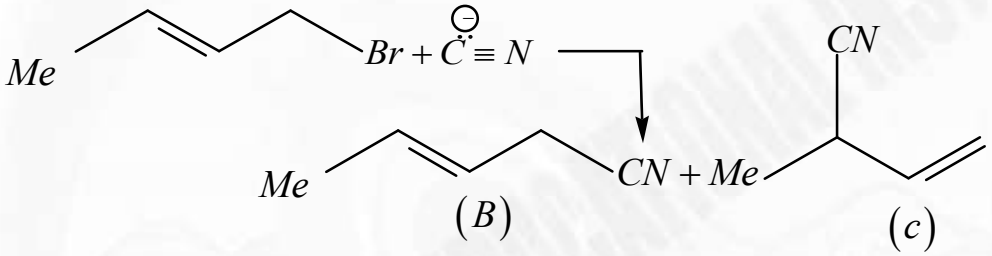
1) If both statement-I and statement-II are correct

2) If both statement-I and statement-II are incorrect

3) If statement-I is correct but statement-II is incorrect

4) If statement-I is incorrect but statement-II is correct

68. Assertion (A):



Reason(R): The product (B) results by SN^2 mechanism and product (C) results by SN^{2I} mechanism

1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

3) (A) True but (R) is false

4) (A) is false but (R) is true

69. Assertion (A): For measuring resistance of an ionic solution an Ac source is used

Reason (R): Concentration of ionic solution will change if DC source is used

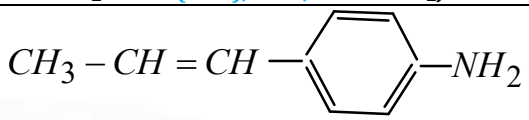
1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

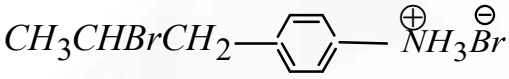
2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

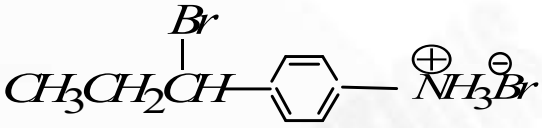
3) (A) True but (R) is false

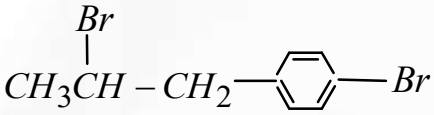
4) (A) is false but (R) is true

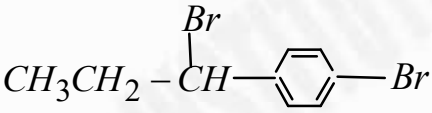
Sri Chaitanya Educational Institutions
Infinity Learn
41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

70. The major product of the reaction of  With HBr (Excess)

1) 

2) 

3) 

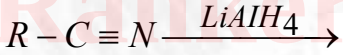
4) 

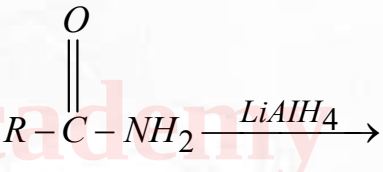
SECTION-II
(NUMERICAL VALUE TYPE

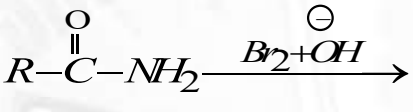
This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

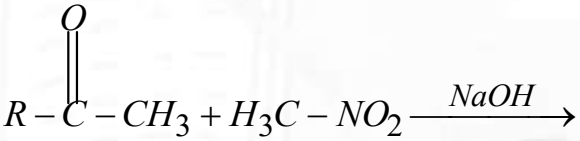
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

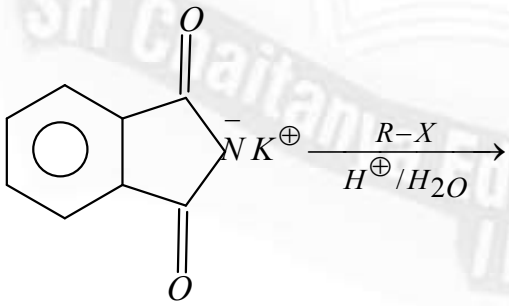
71. Of the following reactions how many reaction, are used for the preparation of amines?

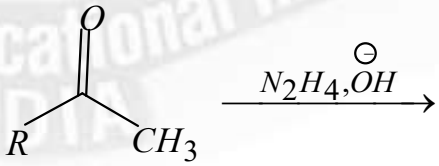
a) 

b) 

c) 

d) 

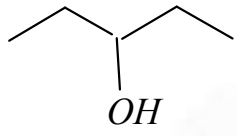
e) 

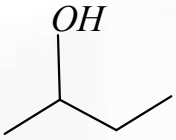
f) 

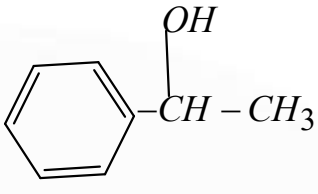
Sri Chaitanya Educational Institutions
Infinity Learn

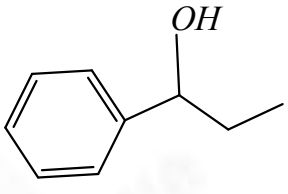

For More Join: @IITJEE_Advanced

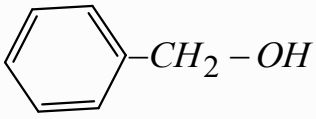
72. Find out number of alcohols that can give positive iodoform test

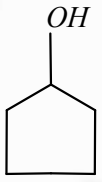
a) 

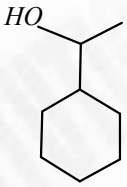
b) 

c) 

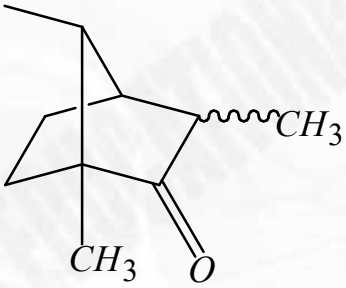
d) 

e) CH_3-CH_2-OH f) 

g) 

h) 

73. The total number of stereoisomers that can exist for M is

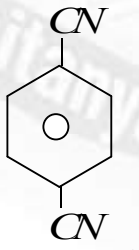


(M)

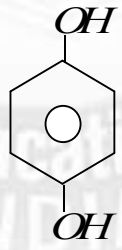
74. H_2S (10 moles) reacts completely with acidified aqueous $KMnO_4$ solution. In this reaction the number of moles of water produced is X and the number of moles of electrons involved is Y . The value of (X+Y) is _____

75. Number of compounds from the following with zero dipole moment____

$HF, H_2, H_2S, CO_2, NH_3, BF_3, CH_4, CHCl_3, SiF_4, H_2O, BeF_2,$



,



Sri Chaitanya Educational Institutions
Infinity Learn
41
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

Sri Chaitanya Educational Institutions | Infinity Learn

41 Years of Excellence 1986-2026

LEADING BY MILES SRI CHAITANYA DOMINATES

JEE ADVANCED 2025

29 Ranks in Top 100 in All-India Open Category



4 Students in Top 11 in JEE-Advanced 2025, All India Open Category

Rankers Academy JEE

16 RANK DEVDUTTA MAJHI HT. No. 255053116*	18 RANK DHARMANA GNANA RUTVIK SAI HT. No. 256055278	19 RANK VANGALA AJAY REDDY HT. No. 256131009	23 RANK AKSH GOGI HT. No. 252071075*	26 RANK P HEMA SAI SURYA KARTHIK HT. No. 256033006	27 RANK SARKARSAMUDRA HT. No. 252071105*
30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048248*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS: 29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS: 113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS: 205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT: 745	NUMBER OF QUALIFIED RANKS: 4,212

0 40 66 15 15 15 | www.srichaitanya.net

For More Join: @IITJEE_Advanced